Lesson Seven

Resource allocation

Software Project Management
5th Edition

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Ch08: Resource allocation

Presented by
Bowen DU

Outline

- Identify the resources which project needs.
- Balance the needs of resources in SDLC.
- Create activity schedule and resource schedule.

Introduction

- In general, the allocation of resources to activities will lead us to review and modify the ideal activity plan. It may cause us to revise stage or project completion dates.
- In any event, it is likely to lead to a narrowing of the time spans within which activities may be scheduled.
- Resourses are always limited.

Introduction

- The final result of resource allocation will normally be a number of schedules
  - An activity schedule: The planned start date and completion date for each activity.
  - A resource schedule: The dates on which each resource will be required and the level of that requirement.
  - A cost schedule: The planned cumulative expenditure incurred by the use of resources over time.

The nature of resources

- In general, resources will fall into one of 7 categories:
  - Labor
  - Equipment
  - Materials
  - Space
  - Services
  - Time
  - Money
  - Entrepreneur

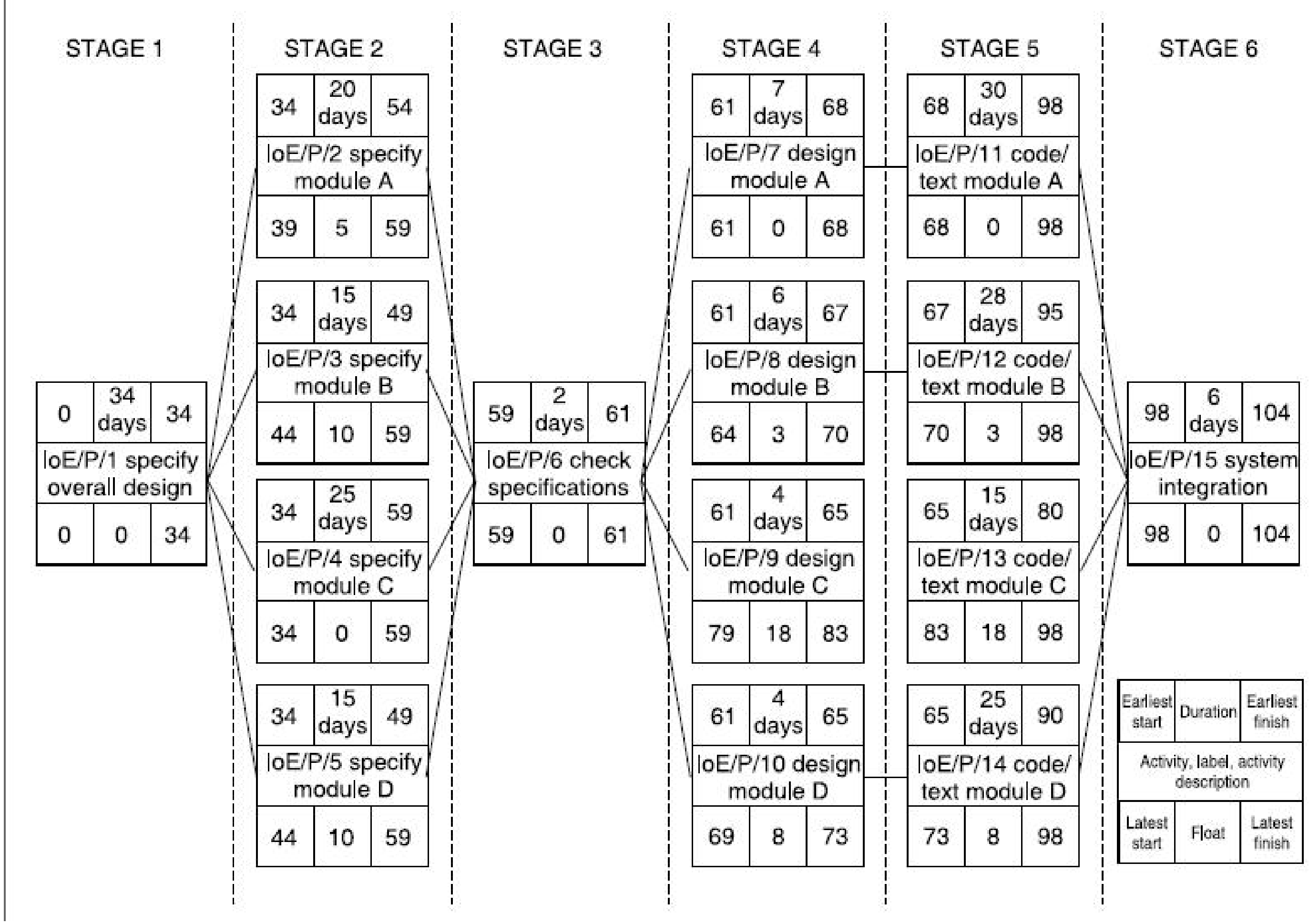
Identifying resource requirements

- The First step in producing a resource allocation plan is to list the resources that will be required along with the expected level of demand.
- This will normally be done by considering each activity in turn and identifying the resources required. However, there will also be resources required that are not activity specific, but are part of the project's infrastructure or required to support other resources (space, or might be required to software engineers).

Case: identifying resource requirements

- Amanda has produced a precedence network for the IOE project and use this as a basis for a resource requirements list.
- Notice:
  - At this stage, she has not allocated individuals to tasks but has decided on the type of staff that will be required.
  - The activity durations assume that they will be carried out by “standard” analysts or software developers.

Case: identifying resource requirements



Case: identifying resource requirements

Stage	Activity	Resource	Days	Quantity	Notes
ALL	Project manager		104 F/T		
1	All	Workstation	–	34	Check software availability
	IoE/P/1	Senior analyst	34 F/T		
2	All	Workstation	–	3	One per person essential
	IoE/P/2	Analyst/designer	20 F/T		
	IoE/P/3	Analyst/designer	15 F/T		

(Contd)

Case: identifying resource requirements

	IoE/P/4	Analyst/designer	25 F/T		
	IoE/P/5	Analyst/designer	15 F/T		Could use analyst/ programmer
3	All	Workstation	–	2	
	IoE/P/6	Senior analyst*	2 F/T		
4	All	Workstation	–	3	As stage 2
	IoE/P/7	Analyst/designer	7 F/T		
	IoE/P/8	Analyst/designer	6 F/T		
	IoE/P/9	Analyst/designer	4 F/T		
	IoE/P/10	Analyst/designer	4 F/T		

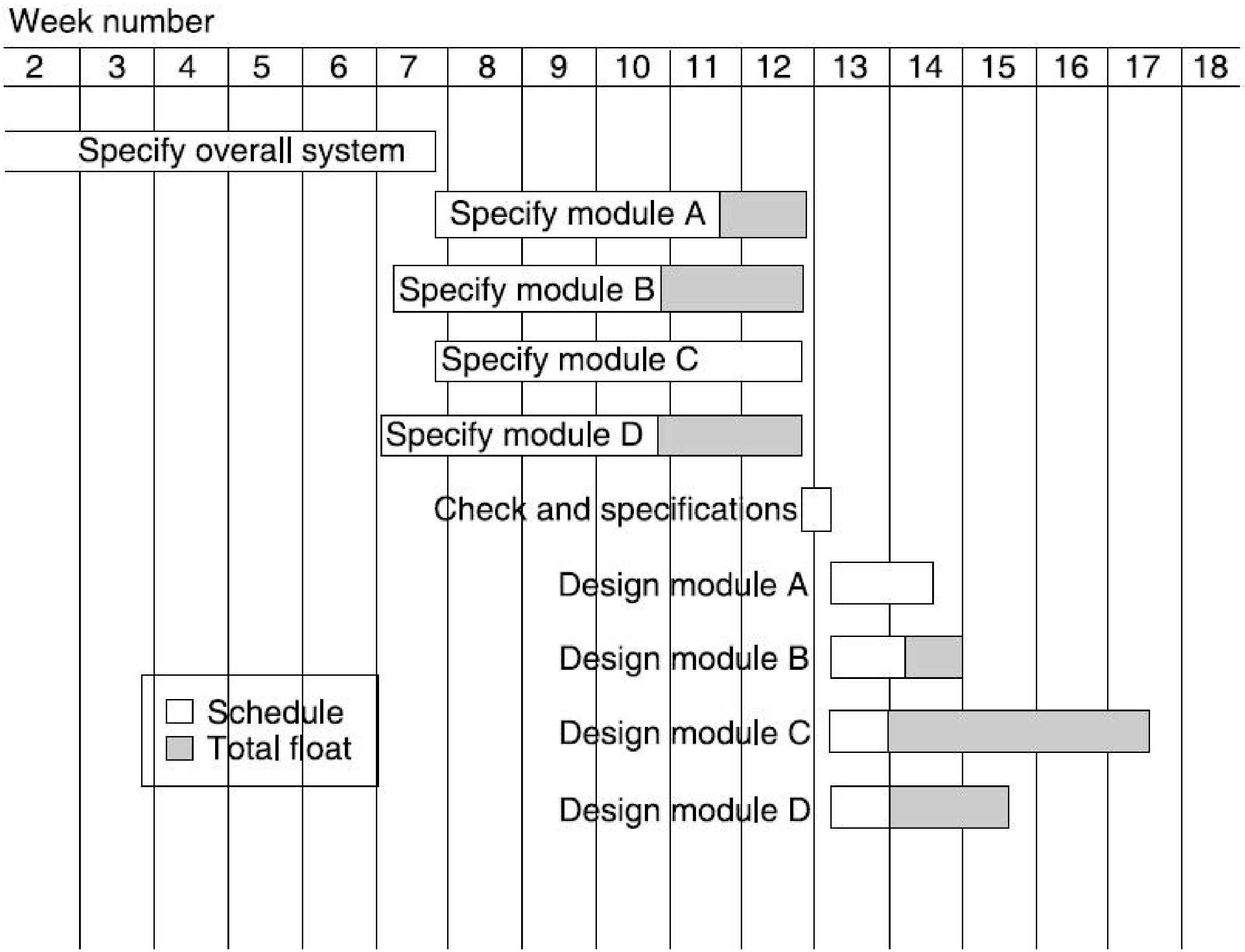
Case: identifying resource requirements

5	All	Workstation	–	4	One per programmer
	All	Office space	–		If contract programmers used
	IoE/P/11	Programmer	30 F/T		
	IoE/P/12	Programmer	28 F/T		
	IoE/P/13	Programmer	15 F/T		
	IoE/P/14	Programmer	25 F/T		
6	All	Full system access	–		Approx. 16 hours for full system test
	IoE/P/15	Analyst/designer	6 F/T		

Scheduling resources

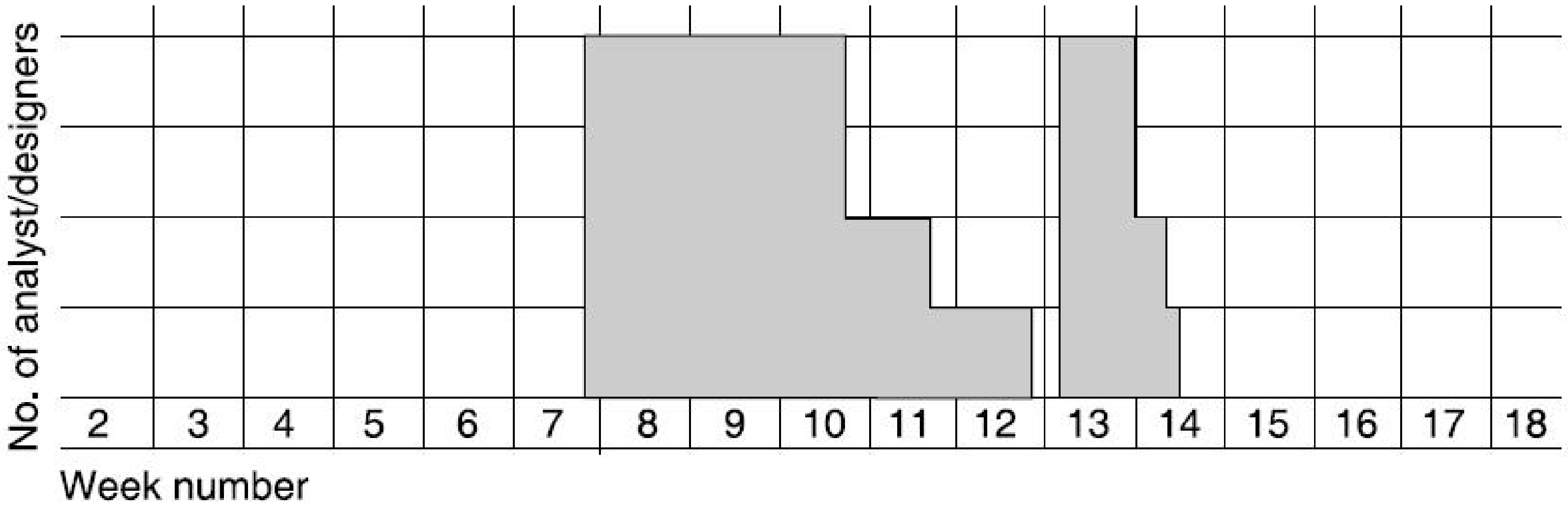
- The Second step is to map the resource requirements produced in the first step on to the activity plan to assess the distribution of resources over the duration of the project.
- This is best done by representing the activity plan as a bar chart and using this to produce a resource histogram for each resource.

Case: scheduling resources



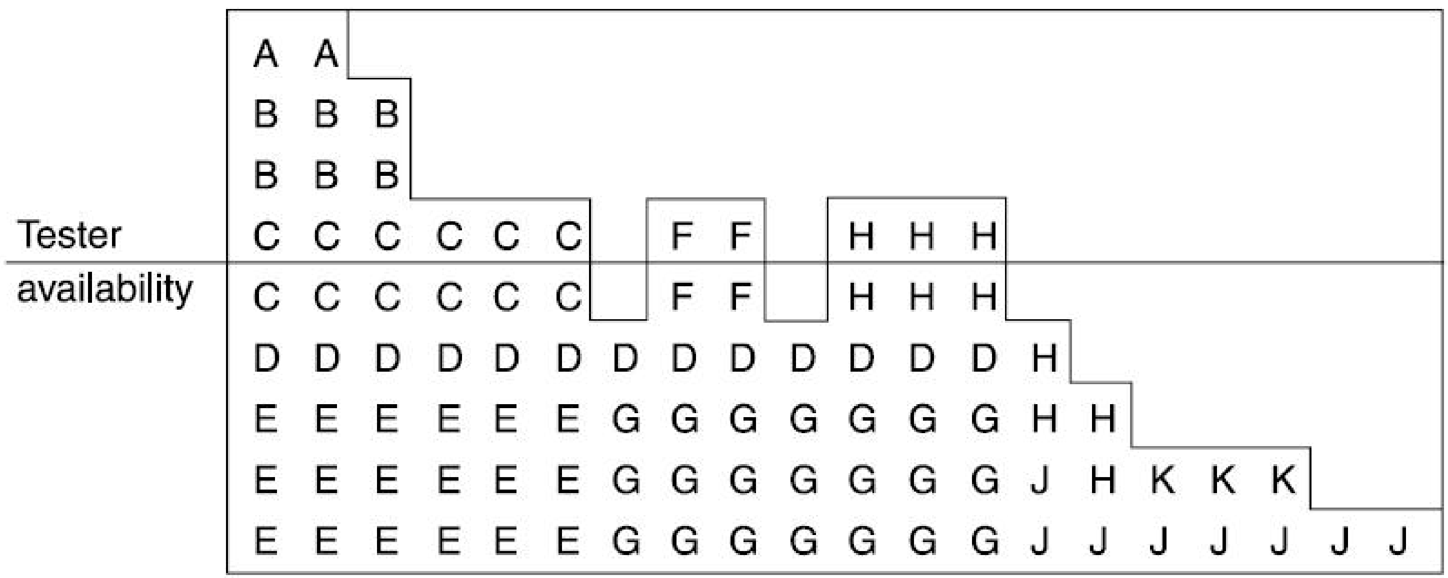
Case: scheduling resources

- The resource histogram poses particular problems in that it calls for two analysts/designers to be idle for 12 days, one for 7 days and one for 2 days between the specification and design stage. This raises the question whether this idle time should be charged to Amanda's project. The ideal resource histogram will be smooth with an initial build-up and a staged run-down.



Case: scheduling resources

- The original histogram was created by scheduling the activities at their earliest start dates. The resource histogram shows the typical peaked shape caused by earliest start date scheduling and calls for a total of 9 staff where only 5 are available for the project.



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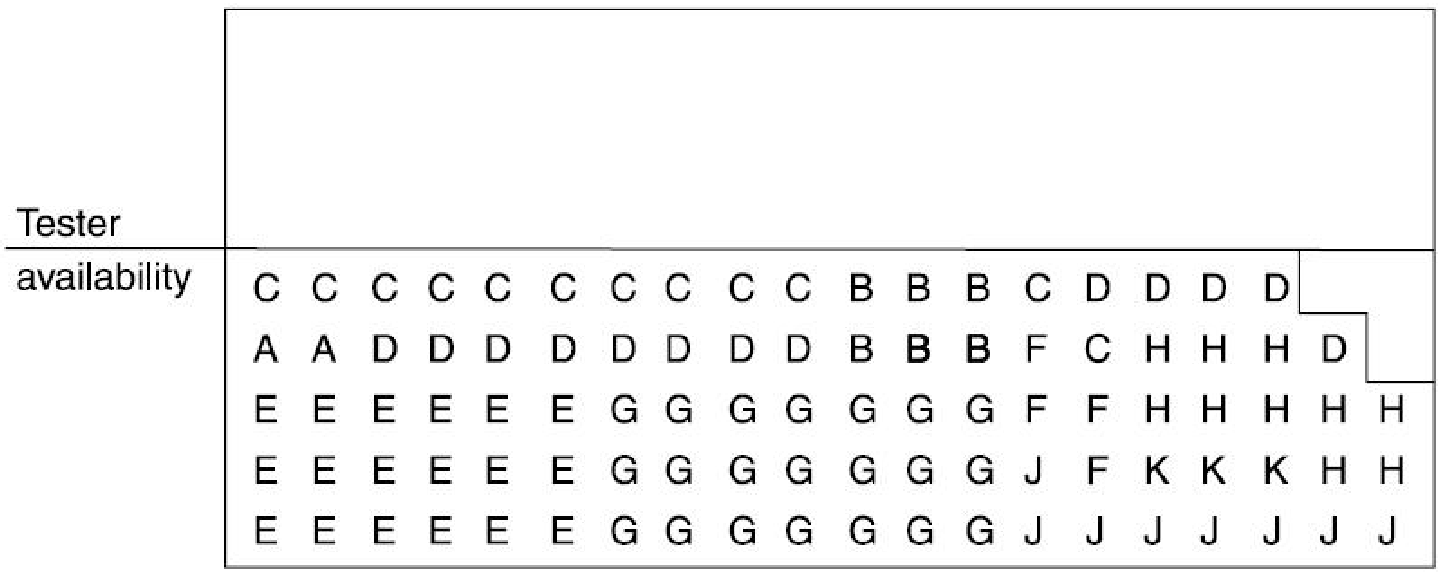


Case: scheduling resources (Cont’s)

- By delaying the start of some of the activities, it has been possible to smooth the histogram and reduce the maximum level of demand for the resource.
- Notice that some activities, such as **C** and **D**, have been split. But in software projects it is difficult to split tasks without increasing the time they take.

Scheduling resources

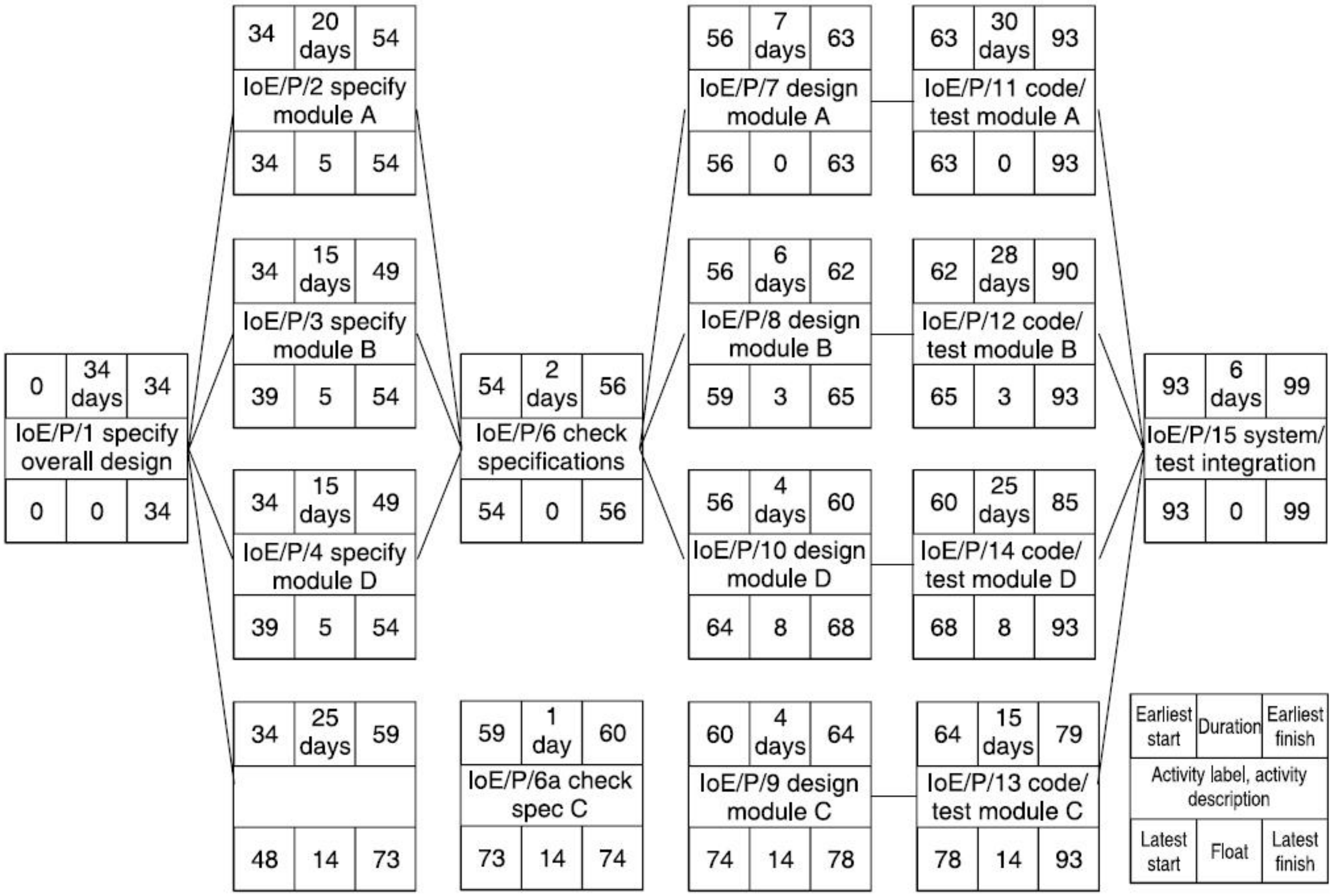
- Amanda has already decided to use only three analyst/designers on the project in order to reduce costs. Her current resource histogram calls for four during both stage2 and stage4.
- Suggest: what might she do to smooth the histogram and reduce the number of analyst/designers required to three.
- It is helpful to prioritize activities so that resources can be allocated to competing activities in some rational order.
- The priority must almost always be to allocate resources to critical path activities and then to those activities that are most likely to affect others.

Scheduling resources

- Two ways of prioritizing activities
  - Total float priority: smallest float has the highest priority.
  - Ordered list priority: according to the criteria.
- Unfortunately, resources smoothing or even containment of resource demand to available levels, is not always possible with planned timescales. Deferring activities to smooth out resource peaks often puts back project completion.
- Sometimes we need to consider ways of increasing the available resource levels or altering working methods.

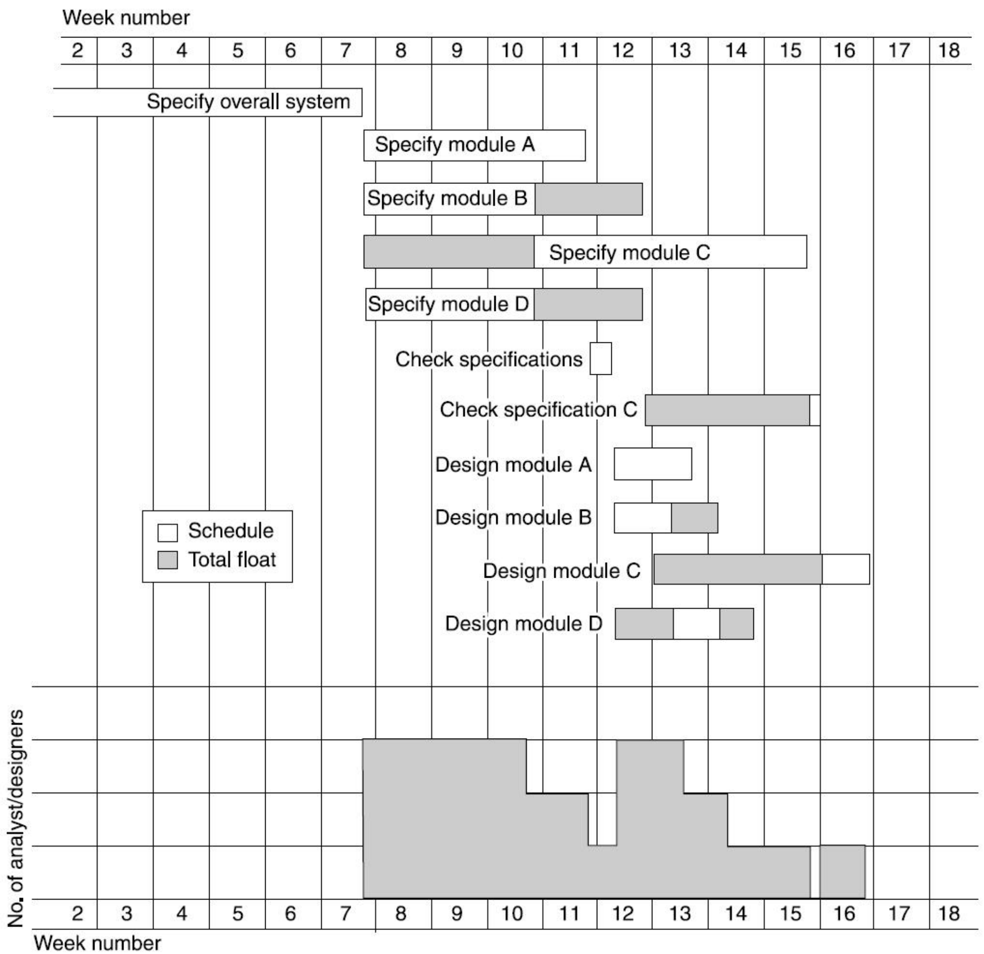
Case: scheduling resources

- Revised precedence network



Case: scheduling resources

- Revised schedule to require **3** analyst/designers



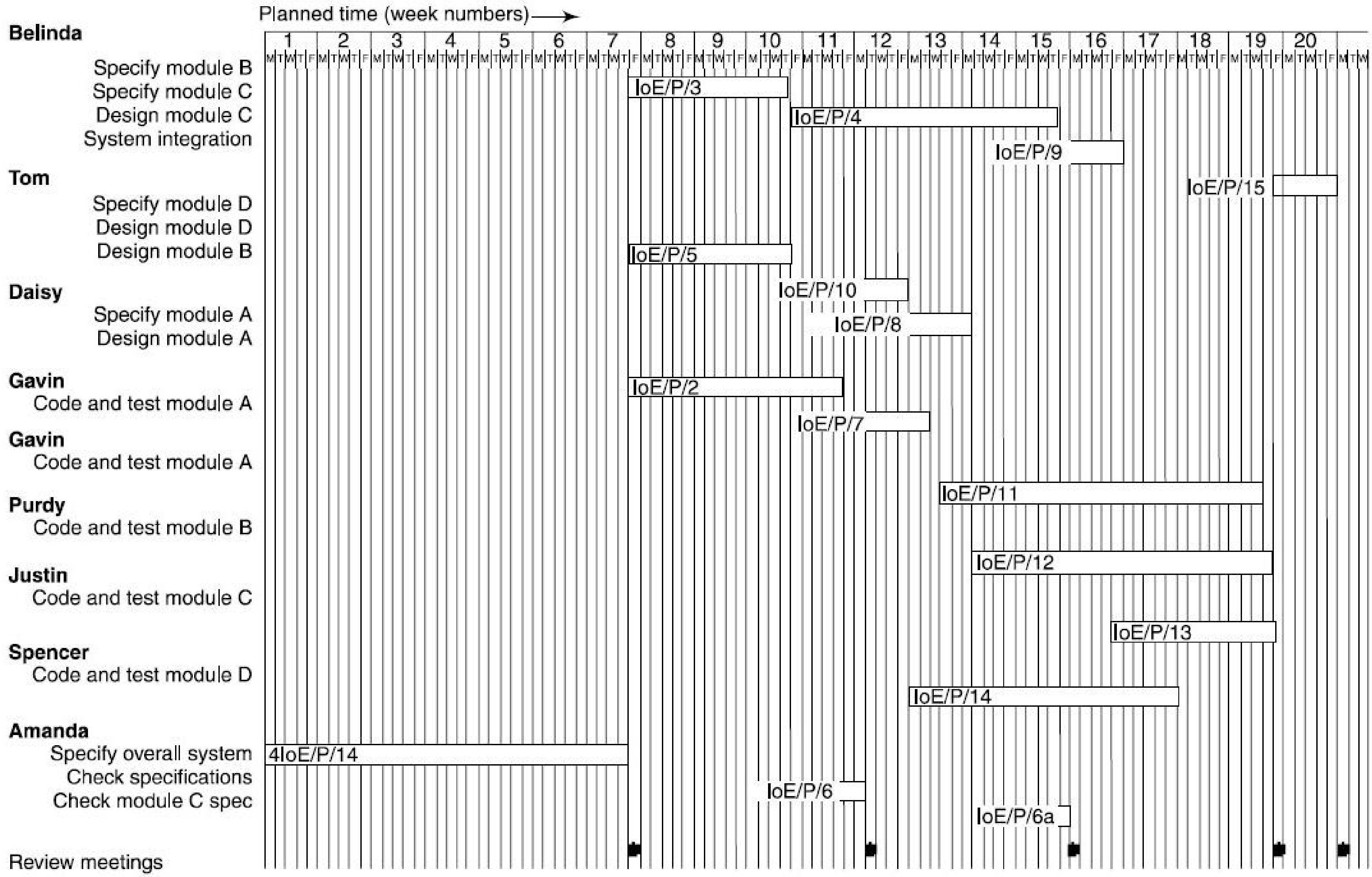
Counting the cost

- Decision making
  - Additional staff ?
  - Lengthening duration of the project ?
- Comparing
  - Cost of employing extra staff.
  - Cost of delayed delivery, increased risk of not meeting the scheduled date.

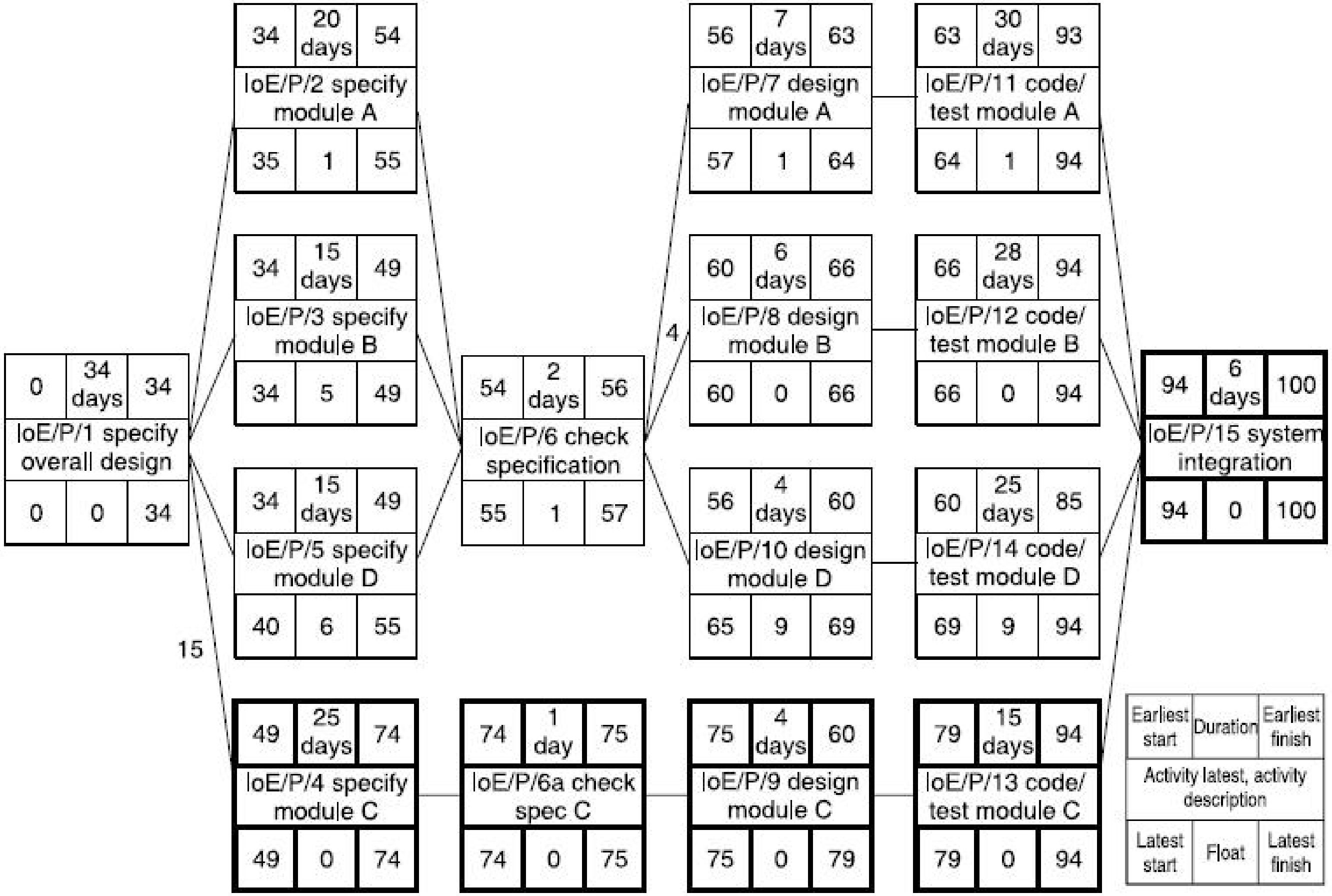
Being specific

- A number of factors need to be thought
  - Availability – a particular individual
  - Criticality - experienced personnel
  - Risk – experienced personnel
  - Training – junior staff
  - Team building – communication, conflict solving etc.

Publish the resource schedule



Revised critical path

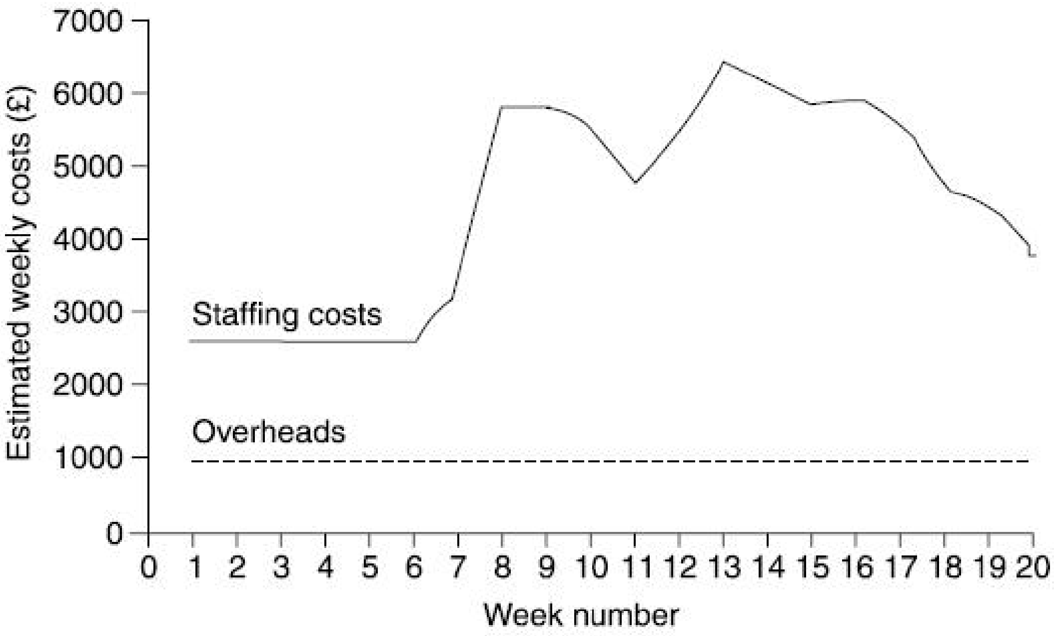


Cost schedules

- It is time to produce a detailed cost schedule showing weekly or monthly costs over the life of the project.
- This will provide a more detailed and accurate estimate of costs and will serve as a plan against which project progress can be monitored.
- In general, costs are categorized as follows
  - Staff cost: staff salaries, other direct and indirect costs.
  - Overhead: expenditure that an organization incurs in the project.
  - Usage charges: resources cost used in the project.

Case: Cost schedules

Staff member	Daily cost (£)
Amanda	300
Belinda	250
Tom	175
Daisy	225
Gavin	150
Purdy	150
Justin	150
Spencer	150



Case: Cost schedules (Cont's)

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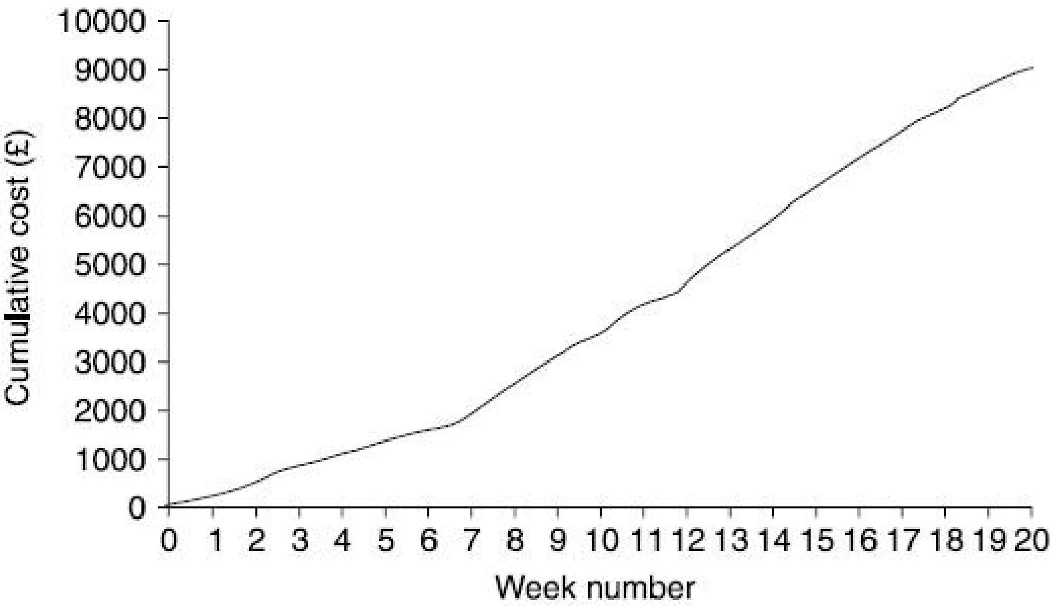

Schedule sequence

- The interplay between the plans and schedules is complex. Any change to any one will affect each of the others.
- Some factors can be directly compared in terms of money – the cost of hiring additional staff can be balanced against the costs of delaying the project's end date.
- Some factors, however, are difficult to express in money terms, for example the cost of an increased risk and will include an element of subjectivity.

Summary of Lesson 7

- In this chapter, we have discussed the problems of allocating resources to project activities and the conversion of an activity plan to a work schedule. In particular, we have seen the importance of the following:
  - Identify all the resources needed.
  - Arrange activity starts to minimize variations in resource levels over the duration of the project.
  - Allocating resources to competing activities in a rational order of priority.
  - Taking care in allocating the right staff to critical activities.

Assignment of Lesson Seven

- Project Charter Peer Review should be completed before 6 April, 2026
- Every team should update the RBS, WBS, estimated working hour for each work, total working hour and schedule before 12 April, 2026
- Everyone should submit his/her own biweekly report (from 30 March to 12 April, one page) to the team leader, which should include the ongoing work and the working hour for work in these two weeks. The team leader should summary the progress of project (one page), according to members' report. Please submit the report at 12th April, 2026